

$$\varepsilon_c - \varepsilon_d \ll \varepsilon_g$$

• 4-dopiran poluprovodnik

•  $n_c(T) = ?$

•  $\mu(T) = ?$

$$n_c(\varepsilon) = \int g(\varepsilon) f(\varepsilon) d\varepsilon$$

$$\hookrightarrow \frac{1}{e^{\beta(\varepsilon - \mu)} - 1}$$

$$n_c = p_c + p_v \text{ (ohranitev nabojov)}$$

$$\hookrightarrow N_d \frac{1}{1 + 2e^{-\beta(\varepsilon_d - \mu)}}$$

Nedegenerirani polpr.

$$n_c = N_c e^{-\beta(\varepsilon_c - \mu)}, N_c = \frac{1}{4} \left( \frac{2m_c^* k_B T}{\pi \hbar^2} \right)^{3/2}$$

$$p_v = p_v e^{\beta(\mu - \varepsilon_v)}, p_v = \frac{1}{4} \left( \frac{2m_v^* k_B T}{\pi \hbar^2} \right)^{3/2}$$

poluprovodnik nedegeneriran:  $\varepsilon_c - \mu \gg k_B T$

$$\mu - \varepsilon_v \gg k_B T \Rightarrow p_v \rightarrow 0$$

$$n_c = N_c e^{-\beta(\varepsilon_c - \mu)}, N_c = \frac{1}{4} \left( \frac{2m_c^* k_B T}{\pi \hbar^2} \right)^{3/2}$$

$$p_v = p_v e^{-\beta(\mu - \varepsilon_v)}, p_v = \frac{1}{4} \left( \frac{2m_v^* k_B T}{\pi \hbar^2} \right)^{3/2}$$

$$\rightarrow N_c e^{-\beta(\varepsilon_c - \mu)} = N_d \frac{1}{1 + 2e^{-\beta(\varepsilon_d - \mu)}}$$

$$\therefore e^{\beta\mu} = y$$

$$y_{1,2} = \frac{-N_c e^{-\beta\varepsilon_c} \pm \sqrt{N_c^2 e^{-2\beta\varepsilon_c} + 8N_c e^{\beta(\varepsilon_c + \varepsilon_d)} N_d}}{4N_d e^{-\beta(\varepsilon_c + \varepsilon_d)}}$$

$$e^{\beta\mu} = \frac{1}{4} e^{\beta\varepsilon_d} \left[ -1 + \sqrt{1 + 8 \frac{N_d}{N_c} e^{\beta(\varepsilon_c - \varepsilon_d)}} \right]$$

A

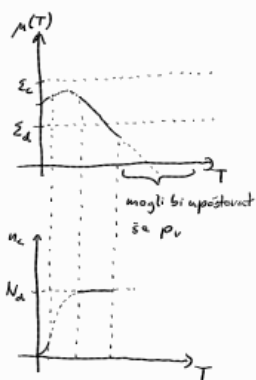
$A \gg 1 \rightarrow \text{nizka } T$

$A \ll 1 \rightarrow \text{visoka } T$

:

$$\mu(T) = \frac{\varepsilon_c + \varepsilon_d}{2} - \frac{3}{4} k_B T \ln T + \dots$$

$$\mu(T) = \varepsilon_c - \frac{3}{2} k_B T \ln T$$



moogli bi upoštevali

š e  $p_v$

① Nizka T

$$n_c = N_c e^{-\beta(\varepsilon_c - \mu)} = \sqrt{\frac{N_d N_c}{2}} e^{-\beta \frac{\varepsilon_c - \varepsilon_d}{2}} \propto T^{3/4} e^{-\frac{\varepsilon_c - \varepsilon_d}{2k_B T}}$$

② Visoka T

$$e^{\beta\mu} = \frac{N_d}{N_c} e^{\beta\varepsilon_c}$$

$$n_c = \dots = N_d$$